Computer Systems I — Final Examination (Reconstructed by GPT)

Part I. Multiple Choice Questions

1. Multiplier Design

A combinational multiplier multiplies two unsigned 5-bit binary numbers.

What is the maximum number of bits required to represent the product?

A. 5 bits
B. 8 bits
C. 9 bits
D. 10 bits

2. BCD Representation

What is the Binary-Coded Decimal (BCD) representation of decimal number 34?

A. 00110100
B. 0011 0100
C. 100010
D. 0010 0010

3. Two's Complement

What is the 8-bit two's complement representation of $(-35)_{10}$?

A. 11011101
B. 11100011
C. 11011100
D. 00100011

4. Boolean Algebra Simplification

Simplify the Boolean function:

$$F = A\overline{C} + ABC + AC\overline{D} + CD$$

Which of the following is equivalent to F?

A. $A + CD$
B. $A\overline{C} + AC + CD$
C. $A + AC\overline{D} + CD$
D. $A\overline{D} + CD$

5. Gate Cost

Consider the Boolean function:

$$F = AB\overline{D} + B + C(B + \overline{A}D)$$

Assume the following gate cost model:

- Each NOT gate costs 1
- Each n-input logic gate costs n

What is the total gate cost of the circuit implementation?

A. 10
B. 12
C. 14
D. 16

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6. Latch Invalid State

An SR latch implemented using NOR gates enters an invalid state under which input condition?

A. $S = 0, R = 0$
B. $S = 0, R = 1$
C. $S = 1, R = 0$
D. $S = 1, R = 1$

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7. Verilog Conditional Operator

The logic function

$$O = \overline{S_0 S_1} I_3 + \overline{S_0} S_1 I_2 + S_0 \overline{S_1} I_1 + S_0 S_1 I_0$$

is equivalent to which Verilog expression using conditional (`?:`) operators?

A.

```
assign O = S0 ? (S1 ? I0 : I1) : (S1 ? I2 : I3);
```

B.

```
assign O = S1 ? (S0 ? I0 : I2) : (S0 ? I1 : I3);
```

C.

```
assign O = S0 & S1 ? I0 : I3;
```

D.

```
assign O = S0 ? I0 : I1;
```

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8. RISC-V Instruction Decode

Which field in an RISC-V R-type instruction specifies the destination register?

A. opcode
B. funct3
C. rd
D. rs1

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9. Sequential Logic Concepts

Which of the following statements is correct?

A. A D flip-flop can be constructed using combinational circuits only.

B. Level-triggered latches usually provide better timing robustness in complex synchronous systems than edge-triggered flip-flops.

C. For the same functionality, a Moore machine generally requires more states than a Mealy machine.

D. None of the above.

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10. RISC-V Instruction Format

The `jalr` instruction in RISC-V belongs to which instruction type?

A. R-type
B. I-type
C. S-type
D. J-type

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11. IEEE 754 Floating Point Representation

What is the IEEE 754 single-precision floating-point representation of -17.5625?

A. 11000001100011000000000000000000

B. 11000001100011010000000000000000

C. 01000001100011010000000000000000

D. 11000001000011010000000000000000

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12. Memory Access

Suppose memory contains the following data:

Memory Address	Stored Data
0x1000	0x1008
0x1004	0x100C
0x1008	0x1004
0x100C	0x100D

Initially, register `R2` contains `0x0004`.

After executing:

```
lw R1, 0x1004(R2)
```

what value will be stored in `R1`?

A. 0x1004
B. 0x1008
C. 0x100C
D. 0x100D

---

13. CPI Calculation

A single-cycle CPU supports four instruction classes: A, B, C, and D.

Their instruction frequencies are:

- A: 10%
- B: 20%
- C: 30%
- D: 40%

The CPI values of A, B, and C are:

- A: 1
- B: 4
- C: 3

After optimization, the CPI of instruction D becomes half of its original value, and the overall CPI becomes 3.

What was the original CPI of instruction D?

A. 4
B. 5
C. 6
D. 8

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14. Single-Cycle vs Multi-Cycle CPU

Which of the following components is necessary in a multi-cycle CPU design but not strictly necessary in a single-cycle CPU?

A. Instruction Register (IR)
B. Program Counter (PC)
C. Instruction Memory
D. Multiplexer (MUX)

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15. Overflow Detection

In two's complement arithmetic, overflow occurs when:

A. Adding a positive number to a negative number.

B. The carry-out bit $c_n = 1$.

C. The sign bits of the operands are different.

D. $c_n \oplus c_{n-1} = 1$.

Part II. Design and Analysis Problems

Problem 1 (10 points)

The Boolean function $Y(A, B, C, D)$ is specified by the following truth table.

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	1
0	0	1	0	X
0	0	1	1	1
0	1	0	0	1
0	1	0	1	X

A	B	C	D	Y
0	1	1	0	0
0	1	1	1	0
1	0	0	0	X
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	X
1	1	1	1	1

where X denotes a don't-care condition.

(a) Canonical Representation (3 points)

Determine:

$$Y = \Sigma m(\cdots)$$

and

$$Y = \Sigma m(\cdots) + \Sigma d(\cdots)$$

where Σd represents the set of don't-care minterms.

(b) Karnaugh Map Minimization (4 points)

Construct the Karnaugh Map for Y.

Identify:

- all Prime Implicants,
- all Essential Prime Implicants,

and derive the minimum SOP (Sum-of-Products) expression.

(c) Minimum POS Form (3 points)

Using the Karnaugh Map, derive the minimum POS (Product-of-Sums) expression of Y.

Problem 2 (10 points)

Given

$$F(A, B, C, D) = \Sigma m(1, 3, 4, 11, 12, 13, 14, 15)$$

Implement F using:

- exactly two 3-to-8 decoders,
- one inverter,
- OR gates with at most four inputs.

Draw the complete logic diagram.

(10 points)

Problem 3 (15 points)

A traffic light controller has three operating states:

State Encoding	Light
00	Green
01	Yellow
11	Red

An input signal X is used to extend the yellow-light duration.

When $X = 1$, the controller remains in the Yellow state.

The system is implemented using two D flip-flops.

(a) State Table

Construct the complete state transition table.

(4 points)

(b) State Diagram

Draw the corresponding state diagram.

(3 points)

(c) Next-State Logic

Using Karnaugh Maps, derive the next-state equations

$$D_1$$

and

$$D_0$$

for the two flip-flops.

(5 points)

(d) Logic Circuit

Draw the complete sequential circuit.

(3 points)

Problem 4 (10 points)

A single-cycle processor supports the following instruction classes.

Instruction Type	Execution Time
R-type	750 ps
Load	1000 ps
Branch	875 ps
I-type	625 ps
Jump	600 ps

Assume all instruction classes occur with equal frequency.

(a)

Determine the clock cycle time and the clock frequency.

(2 points)

(b)

Determine the average CPI.

(2 points)

(c)

Calculate the CPU execution time required to execute

$$10^8$$

instructions.

(3 points)

(d)

Suppose the processor is redesigned so that:

- the execution time from part (c) is reduced to 80% of its original value,
- the total number of clock cycles increases by 20%.

Determine the new clock frequency.

(3 points)

Problem 5 (25 points)

A processor uses a fixed-length 32-bit instruction format.

The ISA contains:

- 130 distinct instructions,
- 64 general-purpose registers.

Every instruction has exactly:

- one destination register field rd,
- one immediate field imm.

All instructions use the same format and occupy exactly one 32-bit word.

(a)

Determine the minimum number of bits required for the opcode field.

(4 points)

(b)

Determine the minimum number of bits required for the destination register field rd.

(4 points)

(c)

Determine the maximum number of bits available for the immediate field.

(4 points)

(d)

Assume the immediate value is interpreted as an unsigned byte address offset for memory access.

What is the maximum memory offset that can be represented?

(5 points)

(e)

Assume the immediate value is represented using two's complement notation.

Determine:

- the largest representable immediate value,
- the smallest representable immediate value.

(8 points)

Problem 6 (20 points)

Consider a processor with separate Instruction Memory and Data Memory.

The processor supports the following three instructions:

Load Word

$$lw\ ri, imm$$

Store Word

$$sw\ ri, imm$$

Add Immediate

$$add\ ri, imm$$

(a)

Complete the interfaces and signal connections of the Register File and Data Memory modules.

(5 points)

(b)

Draw the complete datapath capable of supporting all three instructions.

(8 points)

(c)

Design the control signals required for the datapath.

For each instruction:

- lw
- sw
- add

specify the value of every control signal.

Present your answer in tabular form.

(7 points)